

9	(a)	'light' nuclei combine to form 'heavier' nuclei	B1	[1]
	(b)	(i)		
		<i>either</i> energy = $c^2\Delta m$	C1	
		or energy = $(3.00 \times 10^8)^2 \times 1.66 \times 10^{-27}$	C1	
		energy = $1.494 \times 10^{-10} \text{ J}$		
		= $(1.494 \times 10^{-10}) / (1.60 \times 10^{-13})$	A1	[3]
		= 934 MeV (3 s.f.)		
		(ii)		
		$\Delta m = (2.01356 + 3.01551) - (4.00151 + 1.00867)$		
		= 5.02907 – 5.01018		
		= 0.01889 u	C1	
		energy = 0.01889×934		
		= 17.6 MeV (allow 2 s.f.)	A1	[2]
		(iii)		
		high temperature means high speeds / <u>kinetic</u> energy of nuclei	B1	
		D and T nuclei collide despite repelling one another	B1	[2]

Page 5	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – October/November 2014	9702	42

Section B